VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Practical No. 09

Aim: Develop Decision Tree Classification model for a given dataset and use it to classify a new sample.

Software Required: Python IDE, Anaconda, VS Code, Jupyter Nootbook, Google Colab.

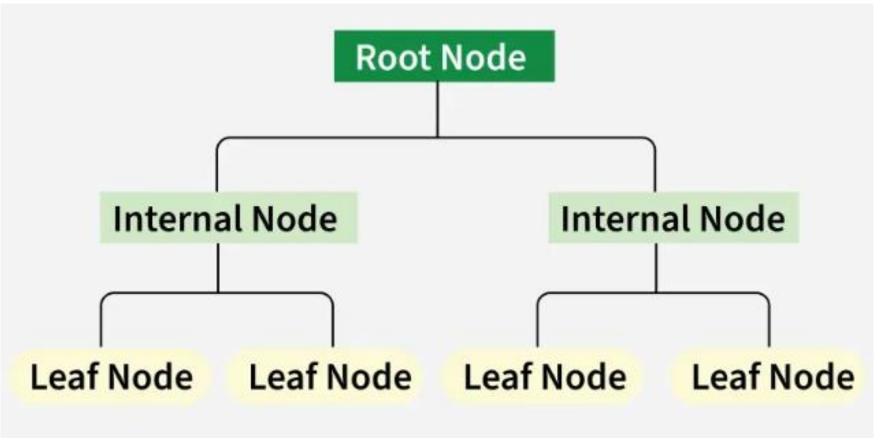
Theory:

Decision tree

Decision tree classification is a supervised machine learning algorithm used to categorize data into predefined classes. It constructs a tree-like model where internal nodes represent tests on features, branches represent the outcomes of these tests, and leaf nodes represent the class labels.

A Decision Tree helps us make decisions by showing different options and how they are related. It has a tree-like structure that starts with one main question called the root node which represents the entire dataset. From there, the tree branches out into different possibilities based on features in the data.

- **Root Node:** Starting point representing the whole dataset.
- **Branches:** Lines connecting nodes showing the flow from one decision to another.
- **Internal Nodes:** Points where decisions are made based on data features.
- **Leaf Nodes:** End points of the tree where the final decision or prediction is made.



A Decision Tree also helps with decision-making by showing possible outcomes clearly. By looking at the "branches" we can quickly compare options and figure out the best choice.

There are mainly two types of Decision Trees based on the target variable:

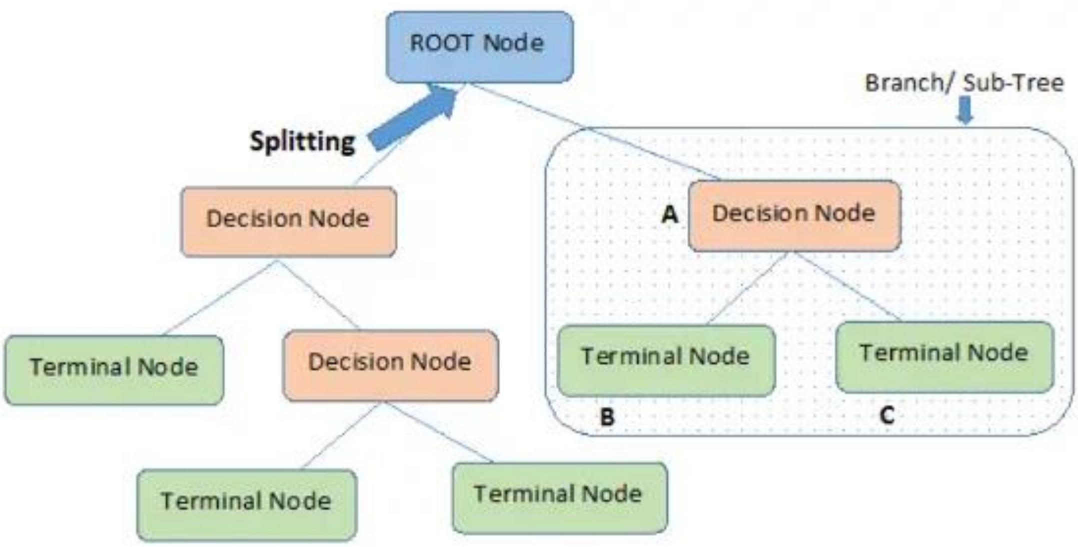
1. **Classification Trees:** Used for predicting categorical outcomes like spam or not spam. These trees split the data based on features to classify data into predefined categories.

2. **Regression Trees:** Used for predicting continuous outcomes like predicting house prices. Instead of assigning categories, it provides numerical predictions based on the input features.

Decision Tree Terminologies

Before learning more about decision trees let's get familiar with some of the terminologies:

- **Root Node**: The initial node at the beginning of a decision tree, where the entire population or dataset starts dividing based on various features or conditions.
- **Decision Nodes**: Nodes resulting from the splitting of root nodes are known as decision nodes. These nodes represent intermediate decisions or conditions within the tree.
- **Leaf Nodes**: Nodes where further splitting is not possible, often indicating the final classification or outcome. Leaf nodes are also referred to as terminal nodes.
- **Sub-Tree**: Similar to a subsection of a graph being called a sub-graph, a sub-section of a these tree is referred to as a sub-tree. It represents a specific portion of the decision tree.
- **Pruning**: The process of removing or cutting down specific nodes in a tree to prevent overfitting and simplify the model.
- **Branch / Sub-Tree**: A subsection of the entire is referred to as a branch or sub-tree. It represents a specific path of decisions and outcomes within the tree.
- **Parent and Child Node**: In a decision tree, a node that is divided into sub-nodes is known as a parent node, and the sub-nodes emerging from it are referred to as child nodes. The parent node represents a decision or condition, while the child nodes represent the potential outcomes or further decisions based on that condition.



How Decision Tree Algorithms Work?

Decision Tree algorithm works in simpler steps:

- **Starting at the Root**: The algorithm begins at the top, called the "root node," representing the entire dataset.
- **Asking the Best Questions:** It looks for the most important feature or question that splits the data into the most distinct groups. This is like asking a question at a fork in the tree.
- **Branching Out**: Based on the answer to that question, it divides the data into smaller subsets, creating new branches. Each branch represents a possible route through the tree.
- **Repeating the Process**: The algorithm continues asking questions and splitting the data at each branch until it reaches the final "leaf nodes," representing the predicted outcomes or classifications.

Advantages of Decision Trees

- **Easy to Understand:** Decision Trees are visual which makes it easy to follow the decision-making process.

- **Versatility**: Can be used for both classification and regression problems.

- **No Need for Feature Scaling**: Unlike many machine learning models, it don't require us to scale or normalize our data.

- **Handles Non-linear Relationships**: It capture complex, non-linear relationships between features and outcomes effectively.

- **Interpretability**: The tree structure is easy to interpret helps in allowing users to understand the reasoning behind each decision.

- **Handles Missing Data**: It can handle missing values by using strategies like assigning the most common value or ignoring missing data during splits.

Disadvantages of Decision Trees

- **Overfitting:** They can overfit the training data if they are too deep which means they memorize the data instead of learning general patterns. This leads to poor performance on unseen data.

- **Instability:** It can be unstable which means that small changes in the data may lead to significant differences in the tree structure and predictions.

- **Bias towards Features with Many Categories:** It can become biased toward features with many distinct values which focuses too much on them and potentially missing other important features which can reduce prediction accuracy.

- **Difficulty in Capturing Complex Interactions:** Decision Trees may struggle to capture complex interactions between features which helps in making them less effective for certain types of data.

- **Computationally Expensive for Large Datasets:** For large datasets, building and pruning a Decision Tree can be computationally intensive, especially as the tree depth increases.

Applications of Decision Trees

Decision Trees are used across various fields due to their simplicity, interpretability and versatility lets see some key applications:

1. **Loan Approval in Banking:** Banks use Decision Trees to assess whether a loan application should be approved. The decision is based on factors like credit score, income, employment status and loan history. This helps predict approval or rejection helps in enabling quick and reliable decisions.

2. **Medical Diagnosis:** In healthcare they assist in diagnosing diseases. For example, they can predict whether a patient has diabetes based on clinical data like glucose levels, BMI and blood pressure. This helps classify patients into diabetic or non-diabetic categories, supporting early diagnosis and treatment.

3. **Predicting Exam Results in Education:** Educational institutions use to predict whether a student will pass or fail based on factors like attendance, study time and past grades. This helps teachers identify at-risk students and offer targeted support.

4. **Customer Churn Prediction:** Companies use Decision Trees to predict whether a customer will leave or stay based on behavior patterns, purchase history, and interactions. This allows businesses to take proactive steps to retain customers.

5. **Fraud Detection:** In finance, Decision Trees are used to detect fraudulent activities, such as credit card fraud. By analyzing past transaction data and patterns, Decision Trees can identify suspicious activities and flag them for further investigation.

Program

```
# Step 1: Import necessary libraries
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
import matplotlib.pyplot as plt

# Step 2: Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Step 3: Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Step 4: Train Decision Tree Classifier
clf = DecisionTreeClassifier(criterion='gini', max_depth=3, random_state=42)
clf.fit(X_train, y_train)
```

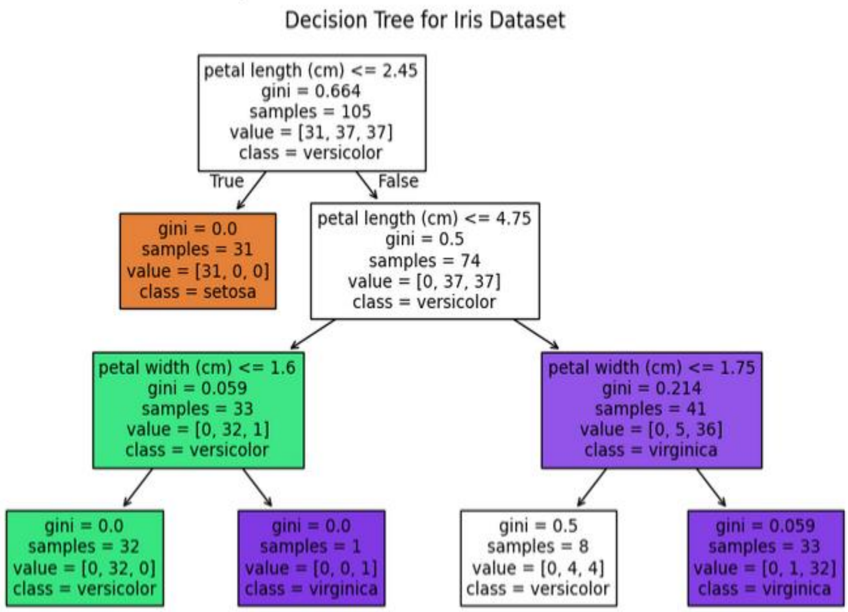
```
# Step 5: Evaluate model
y_pred = clf.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy:.2f}")

# Step 6: Classify new sample
new_sample = [[5.1, 3.5, 1.4, 0.2]]  # Example: one flower's features
predicted_class = clf.predict(new_sample)
print(f"Predicted class for the new sample: {iris.target_names[predicted_class[0]]}")

# Step 7: Visualize the tree
plt.figure(figsize=(10, 6))
plot_tree(clf, filled=True, feature_names=iris.feature_names,
class_names=iris.target_names)
plt.title("Decision Tree for Iris Dataset")
plt.show()
```

Output:

```
Model Accuracy: 1.00
Predicted class for the new sample: setosa
```



Conclusion: By developing a Decision Tree model, we learned how data can be split into rules for classification. The model helps us classify new samples accurately, making it easy to understand and apply in real-world problems.